

MyDesign Portfolio

Element K Initial Template

How to use this template

Students: The details below are commonly found in design project reflections, but the specific assignment given by your instructor may dictate different details. If you make a copy of this template to edit for your own design project reflections, be sure to delete the instructions.

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)
- ideation (element D)
- selection & prototype design (elements D-F)
- prototype building (element G)
- testing (elements H-I)
- evaluation (element J)

For each project step, explain: What went well? What didn't go well? Why?

Elements A-C: Defining the problem

Element A

Honestly this was the second hardest part that we had to deal with in our project.

Throughout researching we struggled to find sources that would match the idea of our project, which was organization in the workplace. Many sources did not relate organization directly to efficacy which needed us to dig deep. We also struggled with finding good quotes to use in the element. The reason why they were so hard to find was that it was harder to do studies on efficacy and relate it to organization due to all the factors that can

affect overall efficacy. On the bright side when putting it all together it went together smoothly and we did that quite well.

Element B

This element went quite well in almost all aspects. We were able to find many designs that were useful along with many different iterations that gave us many different ideas and views on our project. We had no problem dealing with element B at all and got it finished and organized in a concise and timely manner.

Element C

Element C was identifying our design requirements. This part was not too hard and the only problems came through thorough communication with the stakeholder. He often was working and responding was sporadic. When we did get response though we could easily identify the design requirements and write down the necessary information needed.

Ideation: Element D

During this phase we spent quite a bit of time thinking about the different ideas and designs that we could do. Make sketches of almost everything that we could think of. We made many rough sketches to simply put down our ideas. The only problem we had was running out of ideas. During the element we had exhausted our ideas but still needed a few more. Due to the lack of designs out there that fit our design requirements we had a hard time thinking. Throughout the process we overcame this and finally finished up our brainstorming. During this time we also narrowed down the amount of designs that we would be going through to reach our final design.

Selection and prototype design: Elements D

When we finished narrowing down our designs through common sense we reached an important part in our project: the design matrix. When going through the creation of the design matrix we had to do a lot of brainstorming as a team to incorporate our design requirements into a proper design matrix. The hardest part was working with Antonio, the lead stakeholder, to find out what were the most pressing items he wanted in the creation of his socket holder. After putting our designs through a fully completed design matrix we finally started prototyping.

Prototype Building: Element G

This was not a hard time during our process, but by far the longest. When putting together our prototype we had to figure out the most efficient and best way to complete the design. It also needed some time to find out the best dimensions for the socket holder. When we finally got over all of that it was smooth sailing as we completed it within our expected time.

Testing: Elements H-I

This was quite easy as we already had gotten down our tests that we would need to do. The only thing to do was to test the prototype. The first time we had some slight problems, but that was the only problem in the testing procedure. Once we fixed the problem we had reached our final design.

Evaluation: J

When presenting it to multiple stakeholders and experts we got a majority of positive feedback. Only a few personal critiques were mentioned to us and our main stakeholder loved the design and the simplicity. He also found them quite effective in managing the sockets in his shop and he felt like they helped.

Lessons Learned About the Engineering Design Process (K3)

Future engineers creating socket holders which are to be mounted on pegboard should measure far more carefully than ourselves and consider differences in commercially sold peg boards as their products may not fit soundly unless precautions are taken. We learned the importance of these precautions during our many prototypes and failures over the length of the project.